

Nestle Insight

System Data Dictionary and Relational Schema

Final reference for backend, mobile, web, ERD, API design, and testing teams

Prepared from the final agreed system plan

Purpose. This document defines the main data entities, key fields, enums, and relationships for the full Nestle Insight system so the whole team uses the same terminology and structure while building, diagramming, and testing the project.

Scope. The dictionary covers authentication, user and role management, territory and warehouse structure, shops, product master data, stock handling, orders, deliveries, payments, feedback, forecasting inputs, and auditing.

Core Modeling Rules

- The backend is the source of truth. Mobile and web clients do not access the database directly.
- All transactional actions should be traceable through status history or audit logs.
- Territory-based access must be enforced in the backend.
- Rep-created shops should remain operationally usable but marked as provisional until verified.
- Order status and stock movement should never be tracked only in UI state; they must be stored in the database.

Users

Stores all system users.

Field	Type	Description
id	UUID / bigint	Unique user ID
full_name	varchar	User's full name
email	varchar	Login email
phone	varchar	Contact number
password_hash	varchar	Encrypted password
role	enum	Admin, Warehouse Manager, Demand Planner, Sales Representative, Territory Distributor, Shop Owner
status	enum	pending, approved, active, inactive
territory_id	FK	Assigned territory
warehouse_id	FK	Assigned warehouse if relevant
created_at	timestamp	Record created time
updated_at	timestamp	Record updated time

OTP Verifications

Stores OTPs for login and verification.

Field	Type	Description
id	UUID / bigint	Unique OTP record ID
user_id	FK	Related user
otp_code	varchar	Generated OTP
expires_at	timestamp	Expiry time
verified_at	timestamp	Time verified
status	enum	pending, used, expired

Territories

Stores territory details.

Field	Type	Description
id	UUID / bigint	Territory ID
name	varchar	Territory name
region	varchar	Region or province name
status	enum	active, inactive
created_at	timestamp	Created time

Warehouses

Stores warehouse details.

Field	Type	Description
id	UUID / bigint	Warehouse ID
name	varchar	Warehouse name
territory_id	FK	Linked territory
location	text	Address or location
manager_user_id	FK	Warehouse Manager
created_at	timestamp	Created time

Shops

Stores all shops, including rep-assigned shops.

Field	Type	Description
id	UUID / bigint	Shop internal ID
shop_code	varchar	Unique shop code / Shop ID
name	varchar	Shop name
owner_name	varchar	Owner's name
phone	varchar	Shop contact number
address	text	Shop address
territory_id	FK	Linked territory
assigned_sales_rep_id	FK	Assigned Sales Representative

Field	Type	Description
created_by_sales_rep_id	FK	Rep who created it
is_provisional	boolean	Whether shop was field-created and pending verification
has_app_account	boolean	Whether Shop Owner app account exists
status	enum	active, inactive, pending_verification
created_at	timestamp	Created time
updated_at	timestamp	Updated time

Shop Owner Accounts

Links shop owners to shops.

Field	Type	Description
id	UUID / bigint	Record ID
user_id	FK	User account
shop_id	FK	Linked shop
claimed_using_shop_code	boolean	Whether claimed using Shop ID
linked_at	timestamp	Link time

Products

Stores product master data.

Field	Type	Description
id	UUID / bigint	Product ID
sku_code	varchar	SKU code
name	varchar	Product name
category	varchar	Product category
unit_type	varchar	Case, pack, bottle, etc.
price	decimal	Unit price
is_active	boolean	Active status
created_at	timestamp	Created time

Warehouse Stock

Stores stock currently available in warehouse.

Field	Type	Description
id	UUID / bigint	Record ID
warehouse_id	FK	Warehouse
product_id	FK	Product
quantity_available	integer	Available quantity
quantity_reserved	integer	Reserved quantity
updated_at	timestamp	Last updated

Sales Rep Daily Allocations

Stores daily stock allocation for Sales Representatives.

Field	Type	Description
id	UUID / bigint	Allocation ID
sales_rep_id	FK	Sales Representative
warehouse_id	FK	Warehouse
date	date	Allocation date
lorry_number	varchar	Vehicle or lorry number
status	enum	issued, returned, closed
created_at	timestamp	Created time

Sales Rep Allocation Items

Product-level allocation items.

Field	Type	Description
id	UUID / bigint	Record ID
allocation_id	FK	Parent allocation
product_id	FK	Product
issued_quantity	integer	Quantity issued
returned_quantity	integer	Quantity returned
damaged_quantity	integer	Damaged quantity
expired_quantity	integer	Expired quantity

Orders

Stores all orders from Shop Owners or Sales Representatives.

Field	Type	Description
id	UUID / bigint	Order ID
order_number	varchar	Unique order number
shop_id	FK	Shop placing order
created_by_user_id	FK	Shop Owner or Sales Rep
source_type	enum	shop_owner_app, sales_rep
territory_id	FK	Linked territory
warehouse_id	FK	Handling warehouse
payment_method	enum	cash_on_delivery, card
status	enum	created, allocated, in_delivery, completed, closed, failed
total_amount	decimal	Total order value
created_at	timestamp	Order created time
updated_at	timestamp	Updated time

Order Items

Stores products inside an order.

Field	Type	Description
id	UUID / bigint	Record ID
order_id	FK	Related order
product_id	FK	Product
quantity_ordered	integer	Ordered quantity
quantity_delivered	integer	Delivered quantity
quantity_pending	integer	Pending quantity
unit_price	decimal	Product price
subtotal	decimal	Line total

Order Status History

Tracks order state changes.

Field	Type	Description
id	UUID / bigint	Record ID
order_id	FK	Related order
status	enum	Order state
changed_by_user_id	FK	User who changed it
remarks	text	Optional notes
changed_at	timestamp	Change time

Deliveries

Stores delivery records.

Field	Type	Description
id	UUID / bigint	Delivery ID
order_id	FK	Related order
delivery_user_id	FK	Sales Rep or Territory Distributor
delivery_type	enum	sales_rep_delivery, territory_distributor_delivery
territory_id	FK	Territory
status	enum	assigned, in_progress, completed, partial, failed
delivered_at	timestamp	Delivery completion time
remarks	text	Notes

Delivery Items

Stores delivered quantities by product.

Field	Type	Description
id	UUID / bigint	Record ID
delivery_id	FK	Related delivery
product_id	FK	Product
quantity_assigned	integer	Assigned quantity

Field	Type	Description
quantity_delivered	integer	Delivered quantity
quantity_returned	integer	Returned quantity

Product Transactions

Main stock movement log.

Field	Type	Description
id	UUID / bigint	Transaction ID
warehouse_id	FK	Warehouse
product_id	FK	Product
transaction_type	enum	issued, returned, damaged, expired, adjustment
source_role	enum	warehouse_manager, sales_rep, territory_distributor
source_user_id	FK	Related user
reference_type	enum	allocation, delivery, return, manual_adjustment
reference_id	bigint / UUID	Related record ID
quantity	integer	Quantity moved
transaction_date	timestamp	Transaction time
remarks	text	Notes

Cash Collections

Tracks money collected by Territory Distributors or Sales Reps if needed.

Field	Type	Description
id	UUID / bigint	Collection ID
order_id	FK	Related order
collected_by_user_id	FK	User who collected cash
amount	decimal	Amount collected
collection_date	timestamp	Date and time
notes	text	Notes

Payments

Stores payment details.

Field	Type	Description
id	UUID / bigint	Payment ID
order_id	FK	Related order
payment_method	enum	cash_on_delivery, card
payment_status	enum	pending, paid, failed
paid_amount	decimal	Amount paid
paid_at	timestamp	Paid time

Feedback

Optional shop owner feedback.

Field	Type	Description
id	UUID / bigint	Feedback ID
shop_id	FK	Shop
user_id	FK	Shop Owner
order_id	FK	Optional related order
feedback_text	text	Feedback content
created_at	timestamp	Created time

Forecast Inputs / Demand Signals

Used by Demand Planner features.

Field	Type	Description
id	UUID / bigint	Record ID
territory_id	FK	Territory
shop_id	FK	Optional shop
product_id	FK	Product
source_type	enum	order, sales_visit, non_pos_context, ars_import
signal_date	date	Signal date
quantity	integer	Quantity or value used as signal
notes	text	Extra notes

ARS Integration Records

Stores imported or exported ARS-related data.

Field	Type	Description
id	UUID / bigint	Record ID
file_name	varchar	File or import name
imported_by_user_id	FK	User
import_date	timestamp	Import time
status	enum	success, failed, pending
remarks	text	Notes

Audit Logs

Tracks important system actions.

Field	Type	Description
id	UUID / bigint	Audit ID
user_id	FK	User who performed action
action	varchar	Action name

Field	Type	Description
module	varchar	Module name
reference_type	varchar	Related entity type
reference_id	bigint / UUID	Related entity ID
description	text	Action details
created_at	timestamp	Action time

Main Relationships

User Relationships

- One territory can have many users.
- One warehouse can have many warehouse-related users.
- One Sales Representative can assign many shops.
- One Shop Owner user belongs to one shop in the usual case.

Shop Relationships

- One territory has many shops.
- One shop can have many orders.
- One shop may be linked to one Shop Owner account.

Order Relationships

- One order has many order items.
- One order has many status history records.
- One order may have one or more deliveries.
- One order may have one payment.
- One order may have one or more cash collection records.

Inventory Relationships

- One warehouse has many warehouse stock records.
- One Sales Rep allocation has many allocation items.
- One product transaction belongs to one warehouse and one product.

Recommended Enums

UserRole: ADMIN, WAREHOUSE_MANAGER, DEMAND_PLANNER, SALES_REPRESENTATIVE, TERRITORY_DISTRIBUTOR, SHOP_OWNER

UserStatus: PENDING, APPROVED, ACTIVE, INACTIVE

OrderStatus: CREATED, ALLOCATED, IN_DELIVERY, COMPLETED, CLOSED, FAILED

DeliveryStatus: ASSIGNED, IN_PROGRESS, COMPLETED, PARTIAL, FAILED

PaymentMethod: CASH_ON_DELIVERY, CARD

PaymentStatus: PENDING, PAID, FAILED

ProductTransactionType: ISSUED, RETURNED, DAMAGED, EXPIRED, ADJUSTMENT

Relational Schema Summary

Primary keys. Every main table should have a single primary key, preferably UUID or bigint, depending on team preference and Prisma design choice.

Foreign keys. Relations between users, shops, territories, warehouses, orders, deliveries, and transactions must use foreign keys to preserve integrity.

History tables. Order status history and audit logs should be append-only whenever possible.

- users.territory_id -> territories.id
- users.warehouse_id -> warehouses.id
- warehouses.territory_id -> territories.id
- shops.territory_id -> territories.id
- shops.assigned_sales_rep_id -> users.id
- shops.created_by_sales_rep_id -> users.id
- shop_owner_accounts.user_id -> users.id
- shop_owner_accounts.shop_id -> shops.id
- warehouse_stock.warehouse_id -> warehouses.id
- warehouse_stock.product_id -> products.id
- sales_rep_daily_allocations.sales_rep_id -> users.id
- sales_rep_daily_allocations.warehouse_id -> warehouses.id
- sales_rep_allocation_items.allocation_id -> sales_rep_daily_allocations.id
- sales_rep_allocation_items.product_id -> products.id
- orders.shop_id -> shops.id
- orders.created_by_user_id -> users.id
- orders.territory_id -> territories.id
- orders.warehouse_id -> warehouses.id
- order_items.order_id -> orders.id
- order_items.product_id -> products.id
- order_status_history.order_id -> orders.id
- order_status_history.changed_by_user_id -> users.id
- deliveries.order_id -> orders.id
- deliveries.delivery_user_id -> users.id
- deliveries.territory_id -> territories.id
- delivery_items.delivery_id -> deliveries.id
- delivery_items.product_id -> products.id
- product_transactions.warehouse_id -> warehouses.id
- product_transactions.product_id -> products.id
- product_transactions.source_user_id -> users.id
- cash_collections.order_id -> orders.id
- cash_collections.collected_by_user_id -> users.id

- payments.order_id -> orders.id
- feedback.shop_id -> shops.id
- feedback.user_id -> users.id
- feedback.order_id -> orders.id
- forecast_inputs.territory_id -> territories.id
- forecast_inputs.shop_id -> shops.id
- forecast_inputs.product_id -> products.id
- ars_integration_records.imported_by_user_id -> users.id
- audit_logs.user_id -> users.id

Implementation note: This dictionary is a shared reference for the ERD, Prisma schema, API contracts, backend modules, test cases, and diagram work. If the team changes a field name or entity relationship later, update this document first and then update code and diagrams together.